

Competing Risks Survival Analysis: Mortgage Prepayment and Default

IR&C Assignment 3, Part E(i) — Spring 2026

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1. Introduction

Mortgage prepayment (refinancing or sale) and default (foreclosure, short sale) are *competing risks*: the occurrence of one permanently prevents the other. Treating them as a single event — or analyzing each independently with the Kaplan–Meier estimator — biases every probability estimate and conflates two economically opposite processes. This report applies the correct competing-risks framework to 34 million Freddie Mac 30-year fixed-rate mortgages.

The central finding: four major covariates (FICO, LTV, delinquency, investment property) have statistically significant *opposite signs* for prepayment versus default. A combined model misrepresents both mechanisms.

2. Data

Outcome	N	Share
Prepaid (ZBC = 01)	21,970,748	64.6%
Defaulted (ZBC ∈ {02,03,09})	532,563	1.6%
Censored (still active)	11,510,158	33.8%
Total (FRM-360, 1999–2023)	34,013,469	

Covariate snapshots by event type confirm two distinct populations: prepaid loans have higher FICO (+44 vs defaulters), lower LTV (−8.6 pp), and lower origination rate (−85 bp). Cox and CIF models use a **100,000-loan stratified sample** (proportional by vintage year).

3. Two Competing Causes: Exploratory Analysis

4. Aalen–Johansen CIF and the KM Bias

4.1 Why KM fails under competing risks

The Kaplan–Meier estimator treats competing exits as randomly censored, inflating each cause-specific probability:

$$\hat{F}_{\text{KM}}^{(k)}(t) = 1 - \prod_{t_j \leq t} \left(1 - \frac{d_j^{(k)}}{n_j^{\text{KM}}} \right). \quad (1)$$

Defaulted loans — the *least likely to prepay* — are dropped from n_j^{KM} , so every KM increment exceeds the true value. The bias compounds: $\hat{F}_{\text{AJ}}^{(\text{pre})}(t) \leq \hat{F}_{\text{KM}}^{(\text{pre})}(t)$ for all t .

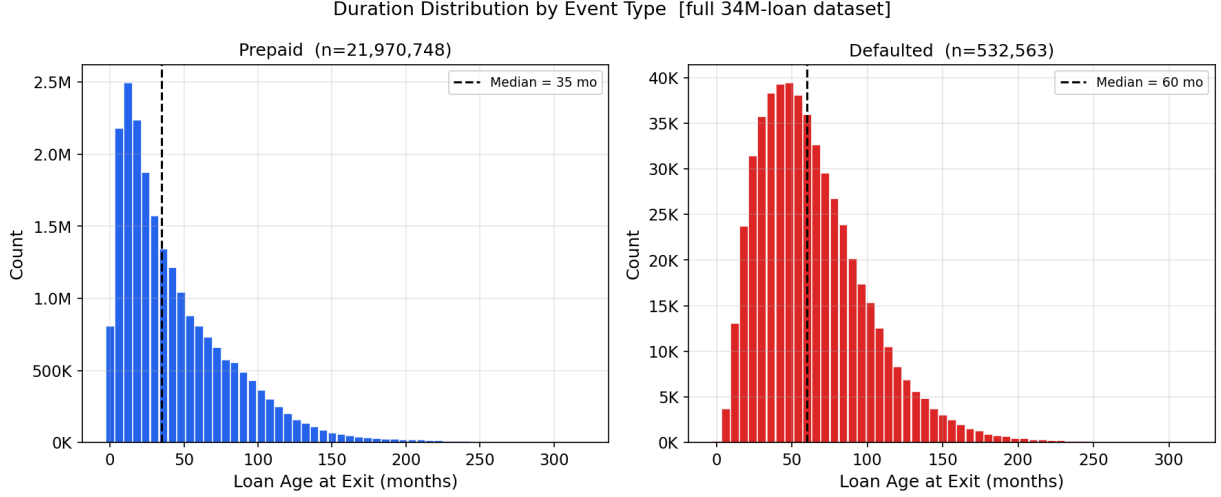


Figure 1: Duration distributions by event type (full 34M dataset). Prepayment peaks at ~ 24 months (rate-option exercise is front-loaded); default peaks at ~ 60 months (gradual financial deterioration). The fundamentally different shapes imply different hazard functions — a single-baseline Cox model cannot represent both simultaneously.

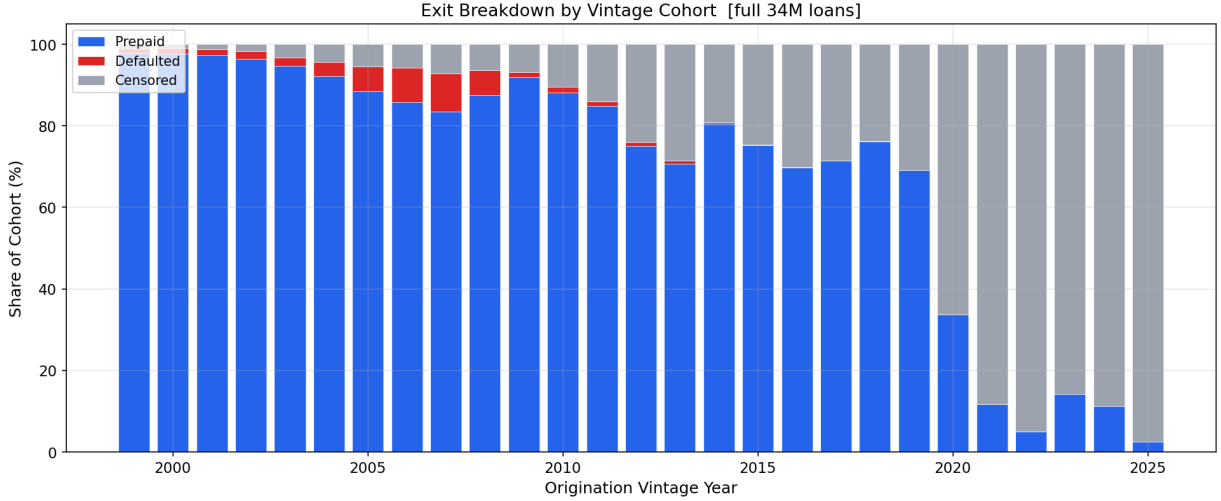


Figure 2: Vintage cohort analysis. The 2007 cohort reaches $\approx 8.5\%$ default — $6\times$ the long-run average of $\approx 1.4\%$ — driven by GFC underwriting. Post-2020 vintages show a large censored share: borrowers who originated at 3% face a strong rate lock-in disincentive to refinance.

4.2 The Aalen–Johansen estimator

$$\hat{F}_k(t) = \sum_{t_j \leq t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j} \quad (2)$$

$\hat{S}(t_j^-)$ uses *all* exits to weight each increment, satisfying the partition identity $\sum_k \hat{F}_k(t) + \hat{S}(t) = 1$ for all t — impossible under KM.

4.3 Results

Horizon	$\hat{F}_{AJ}^{(\text{pre})}$	$\hat{F}_{AJ}^{(\text{def})}$	Still active	KM bias
5 yr	61.4%	1.05%	37.6%	+0.45 pp
10 yr	83.3%	2.23%	14.5%	+1.31 pp
20 yr	91.8%	3.10%	5.1%	+2.23 pp

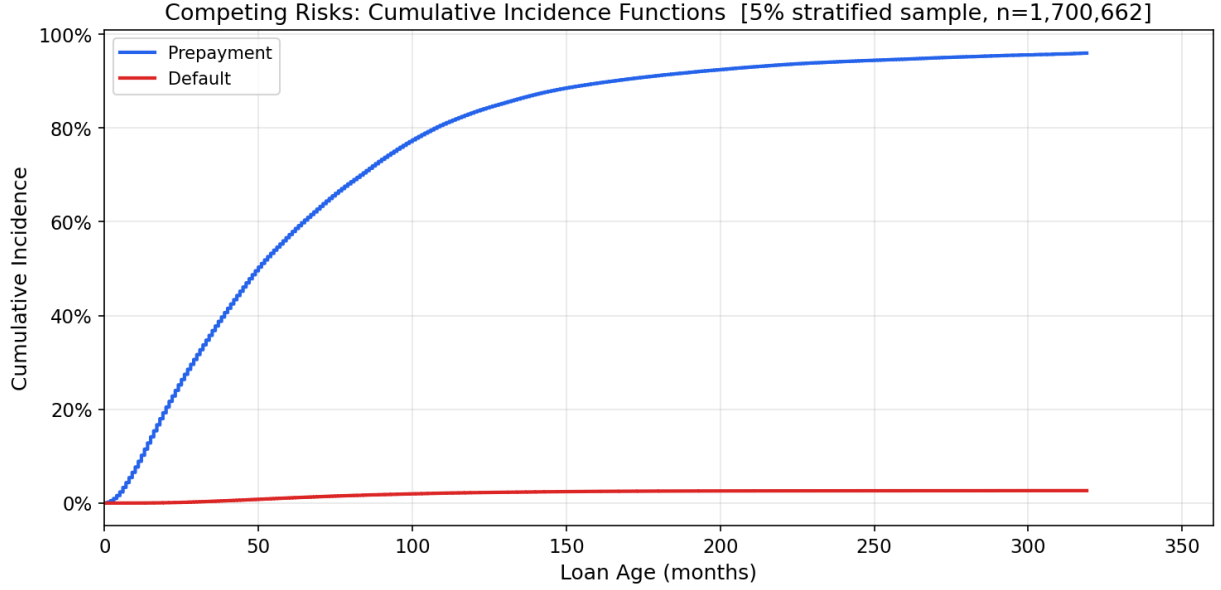


Figure 3: Aalen–Johansen CIF. Prepayment dominates 37:1 at 10 years. The prepayment curve shows burnout: slope decelerates after month ≈ 40 as the most rate-sensitive borrowers exit. Default grows nearly linearly.

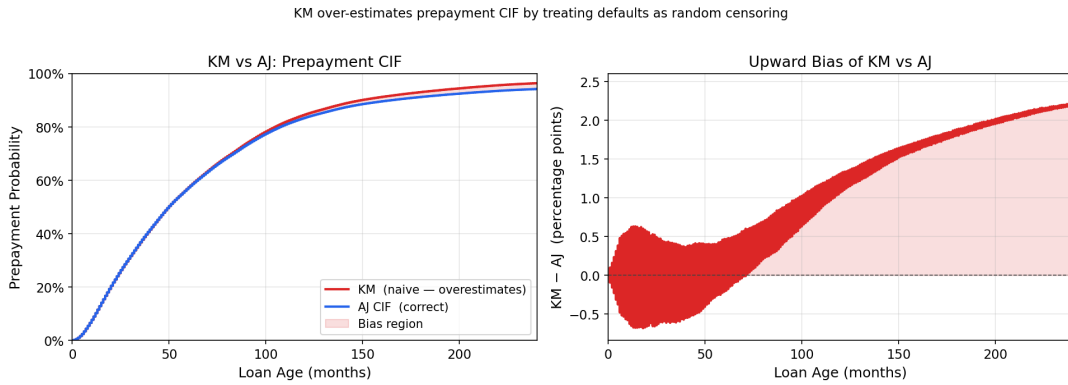


Figure 4: KM vs. AJ prepayment CIF. The gap widens monotonically as more defaulted loans accumulate and are mis-treated as random censoring. A 1 pp over-statement of the 10-year prepayment probability translates to $\approx \$1\text{M}$ of mispriced duration in a $\$100\text{M}$ MBS pool.

5. Risk Heterogeneity: Stratified CIFs

FICO separates default by $6\times$: default CIF ranges from $\approx 1\%$ ($\text{FICO} > 740$) to $\approx 6\%$ ($\text{FICO} < 680$) at 20 years. Prepayment CIFs converge across FICO buckets: FICO enables refinancing *access*, not refinancing *desire*.

LTV doubles default risk (>80 : $\approx 4.5\%$ vs $\approx 2\%$ at 20 yr) while leaving prepayment curves nearly identical. LTV blocks the refinancing path (negative equity) but not exit by sale or payoff.

Vintage era is the dominant stratifier. GFC vintages (1999–2007) reach $\approx 4\%$ default CIF; post-2013 approaches zero. Post-2020 prepayment CIF is nearly flat: rate lock-in.

6. Cause-Specific Cox Regression

Two independent Cox models fit one per cause, treating the alternative as censored:

$$h_k(t \mid \mathbf{x}) = h_{0k}(t) \exp(\boldsymbol{\beta}_k^\top \mathbf{x}), \quad k \in \{\text{pre}, \text{def}\}. \quad (3)$$

Partial likelihood:

$$\ell(\beta_k) = \sum_{i: E_i=k} \left[\beta_k^\top \mathbf{x}_i - \log \sum_{j \in \mathcal{R}(t_i)} e^{\beta_k^\top \mathbf{x}_j} \right]. \quad (4)$$

13 z-scored features; ℓ_2 penalizer = 0.1.

Covariate	Pre $\hat{\beta}$	Def $\hat{\beta}$	Interpretation
Rate incentive	+0.256	+0.021	Refinancing driver; no default content
Mortgage rate	+0.250	+0.060	High orig. rate $\rightarrow$ future refi opportunity
UPB	+0.161	−0.010	Fixed refi cost amortized over larger balance
FICO	+0.010	−0.047	Opposite: access to refi vs. credit quality
LTV	−0.011	+0.041	Opposite: equity enables refi; traps defaulter
Max Delinquency	−0.245	+0.298	Opposite: distressed loans default, skip refi
Inv. property	−0.297	+0.270	Opposite: investors walk rather than refi

The *rate incentive* coefficient ($e^{0.256} \approx 1.29$) captures the classic prepayment S-curve: a 1-SD increase in the rate spread raises prepayment hazard by 29%. Its near-zero default coefficient (+0.021) confirms it carries no credit-risk information.

7. Fine–Gray Subdistribution Hazard

Fine–Gray directly models the CIF by retaining already-defaulted loans in the risk set with IPCW weights $w_i(t) = \hat{G}(t_i)/\hat{G}(\min(t_i, t))$:

$$\hat{F}_k(t) = 1 - \exp\left(-\int_0^t \tilde{h}_{0k}(s) e^{\gamma_k^\top \mathbf{x}} ds\right). \quad (5)$$

The maximum coefficient difference vs. cause-specific Cox is $|\Delta| = 0.004$. With only 1.6% default rate, the IPCW-weighted competing-event borrowers contribute negligibly: $\tilde{h}_1(t) \approx h_1(t)$ when $F_2(t) \rightarrow 0$.

Use case: cause-specific Cox for causal mechanism estimation; Fine–Gray for CIF prediction and MBS loss forecasting.

8. Key Findings and Insights

- Competing risks are not optional.** Four covariates (FICO, LTV, Max Delinquency, Investment property) have opposite signs between prepayment and default. A combined model cancels both signals.
- KM bias reaches 2.2 pp at 20 years.** Systematically over-stating the prepayment probability misprices the embedded duration option in MBS — $\approx \$1\text{M}$ per $\$100\text{M}$ pool at the 10-year horizon.
- Vintage era dominates all other risk stratifiers.** GFC-era (1999–2007) default CIF $\approx 4\%$ vs ≈ 0 post-2013. Post-2020 prepayment CIF is flat: rate lock-in has suspended the refi cycle.
- FICO separates default 6 $\times$; it barely separates prepayment.** FICO belongs in a default model. A high-FICO borrower still needs a rate incentive and sufficient equity to actually refinance.

5. **Rate incentive is the dominant, cause-pure prepayment signal.** $\hat{\beta} = +0.256$ ($e^{0.256} \approx 1.29$) for prepayment; ≈ 0 for default. All rate-shock scenario analysis should route through this coefficient.
6. **Fine–Gray $\approx$ cause-specific Cox in this dataset; the distinction matters in crisis pools.** With 1.6% default, IPCW weights are near 1. In subprime or GFC-era pools with high competing-event rates, Fine–Gray would give materially different CIF forecasts.

Priority improvements: (1) Replace static “Max Delinquency” with a rolling 3-month delinquency indicator. (2) Add an $LTV \times \text{rate_incentive}$ interaction: high-LTV borrowers cannot refi even when incentivized. (3) Stratify $h_{0k}(t)$ by vintage era to capture cohort-specific burnout patterns.

References

- [1] Fine, J. P. and Gray, R. J. (1999). *JASA* 94:496–509.
- [2] Aalen, O. O. and Johansen, S. (1978). *Scand. J. Statistics* 5:141–150.
- [3] Sadhwani, A., Giesecke, K. and Sirignano, J. (2021). *J. Financial Econometrics* 19:313–368.

Stratified Competing-Risk CIFs [5% sample]

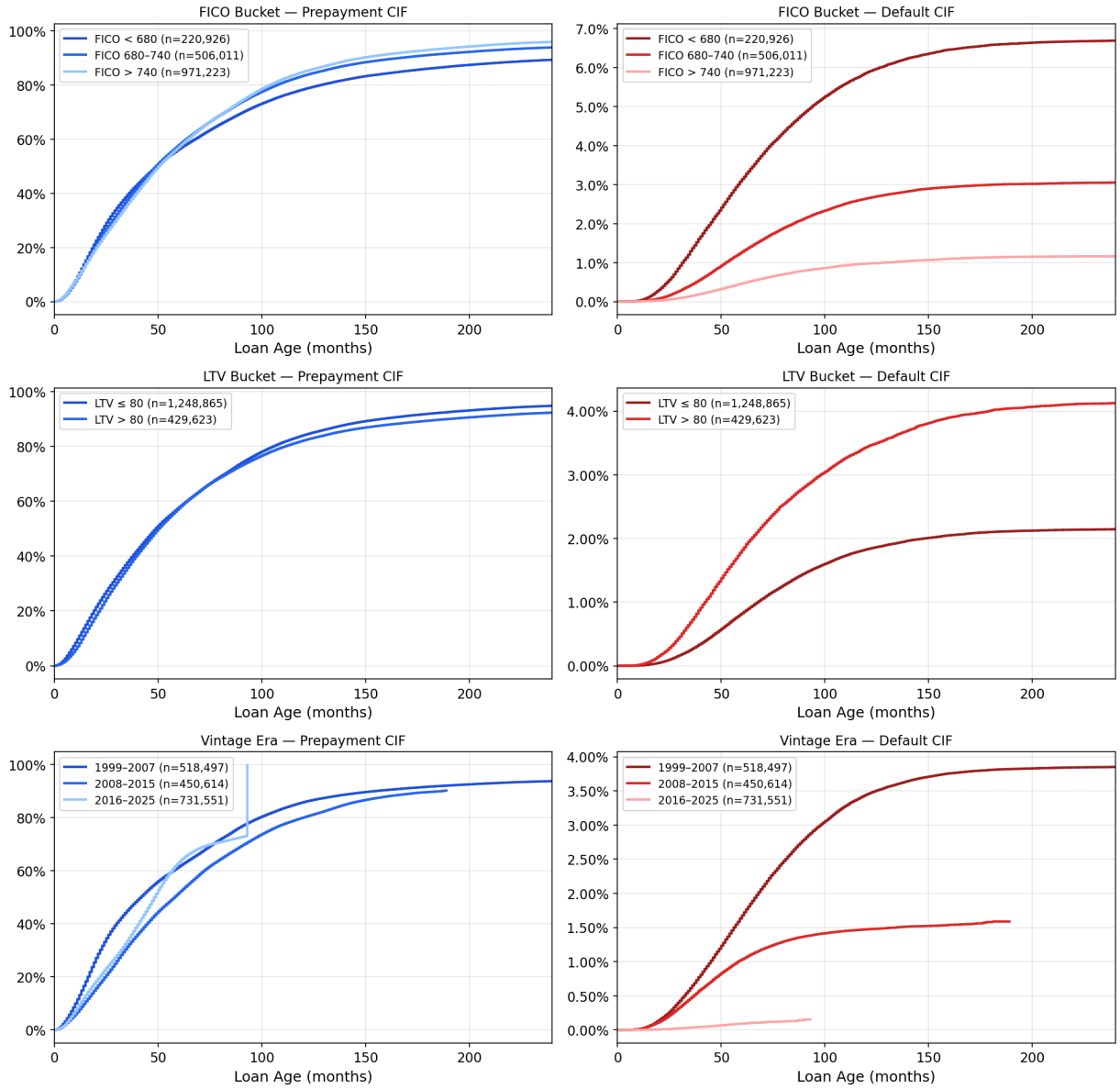


Figure 5: Aalen-Johansen CIF by FICO (top), LTV (middle), and vintage era (bottom). Left: prepayment. Right: default.

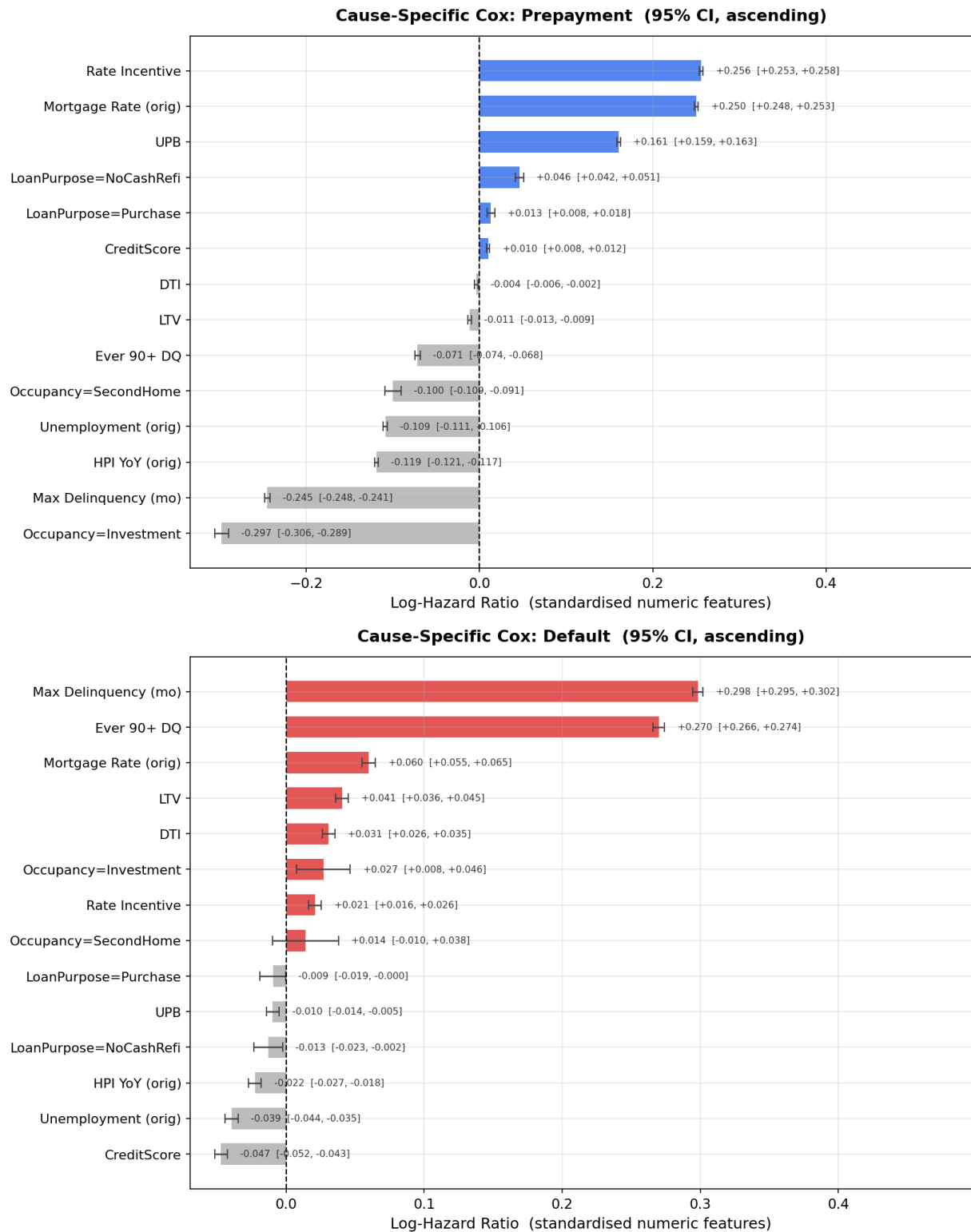


Figure 6: Cause-specific Cox forest plots. Top: prepayment model. Bottom: default model. Note the sign reversals across the two panels.